NOVA SCHOOL OF SCIENCE & TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE

GONÇALO GOMES RODRIGUES

Master in Software Engineering Degree

USE OF DIFFERENTIAL PRIVACY FOR UNOBSERVABLE PRIVACY-PRESERVED COMMUNICATION

SOME THOUGHTS ON THE LIFE, THE UNIVERSE, AND EVERYTHING ELSE

MASTER IN COMPUTER SCIENCE AND ENGINEERING

NOVA University Lisbon
month, year

NOVA SCHOOL OF SCIENCE & TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE

USE OF DIFFERENTIAL PRIVACY FOR UNOBSERVABLE PRIVACY-PRESERVED COMMUNICATION

SOME THOUGHTS ON THE LIFE, THE UNIVERSE, AND EVERYTHING ELSE

GONÇALO GOMES RODRIGUES

Master in Software Engineering Degree

Adviser: Henrique João
Full Professor, NOVA University Lisbon

Co-advisers: João Vilalonga
Associate Professor, NOVA University Lisbon

John Doe other Co-Adviser Name
Full Professor, NOVA University Lisbon

Examination Committee

Chair: Name of the committee chairperson
Full Professor, FCT-NOVA

Rapporteur: Name of a rapporteur
Associate Professor, Another University

Members: Another member of the committee
Full Professor, Another University

Yet another member of the committee
Assistant Professor, Another University

MASTER IN COMPUTER SCIENCE AND ENGINEERING

NOVA University Lisbon
month, year

Use of Differential Privacy for Unobservable Privacy-Preserved Communication
Some thoughts on the Life, the Universe, and Everything Else

Copyright © Gonçalo Gomes Rodrigues, NOVA School of Science and Technology, NOVA University Lisbon.
The NOVA School of Science and Technology and the NOVA University Lisbon have the right, perpetual and without geographical boundaries, to file and publish this dissertation through printed copies reproduced on paper or on digital form, or by any other means known or that may be invented, and to disseminate through scientific repositories and admit its copying and distribution for non-commercial, educational or research purposes, as long as credit is given to the author and editor.

This document was created with the (pdf/Xe/Lua)LaTeX processor and the NOVAthesis template (v7.1.29) [23].

Dedicatory lorem ipsum.

Acknowledgements

Acknowledgments are personal text and should be a free expression of the author.

However, without any intention of conditioning the form or content of this text, I would like to add that it usually starts with academic thanks (instructors, etc.); then institutional thanks (Research Center, Department, Faculty, University, FCT / MEC scholarships, etc.) and, finally, the personal ones (friends, family, etc.).

But I insist that there are no fixed rules for this text, and it must, above all, express what the author feels.

"You cannot teach a man anything; you can only help him discover it in himself."

— **Galileo**, Somewhere in a book or speach
(Astronomer, physicist and engineer)

Abstract

We connect to the internet daily, gaining access to an overwhelming flow of information about the world. Freedom of speech is an essential right, assumed as an unquestionable part of our daily lives. However, this right is denied by authoritarian regimes and entities that control the flow of information, in order to maintain power and control over their citizens. These forces have continuously advanced their methods and resources to enforce such oppressive policies more effectively. Therefore, it is crucial to ensure that those living under such oppression gain access to the tools and content denied to them and have the means to communicate freely, securely, and without fear.

To tackle this challenge, we aim to develop a communication network, on top of Tor network, capable of countering state-of-the-art correlation attacks to ensure user anonymity and privacy. By incorporating Differential Privacy principles, the solution introduces carefully calibrated noise into traffic flow and ensures robust and formally proven privacy guarantees, distinguishing itself from solutions that rely on practical testing. Furthermore, Differential Privacy also provides customizable and adjustable privacy controls, empowering users to balance the trade-off between privacy and utility to their needs.

Keywords: Anonymity Networks, Differential Privacy, Privacy-Preserving Communication, Covert Communication, Traffic Analysis Attacks

Resumo

Ligamo-nos diariamente à Internet, tendo acesso a um fluxo avassalador de informações sobre o mundo. A liberdade de expressão é um direito essencial, assumido como parte inquestionável do nosso quotidiano. No entanto, este direito é negado por regimes autoritários e entidades que controlam o fluxo de informação, a fim de manter o poder e o controlo sobre os seus cidadãos. Estas forças têm vindo a desenvolver continuamente os seus métodos e recursos para aplicar essas políticas opressivas de forma mais eficaz. Por essa razão, é crucial garantir que aqueles que vivem sob tal opressão tenham acesso às ferramentas e conteúdos que lhes são negados e disponham dos meios para comunicar livremente, em segurança e sem medo.

Para fazer face a este desafio, pretendemos desenvolver uma rede de comunicação, sobre a rede Tor, capaz de contrariar ataques de correlação de última geração para garantir o anonimato e a privacidade dos utilizadores. Ao incorporar os princípios da Privacidade Diferencial, a solução introduz ruído cuidadosamente calibrado no fluxo de tráfego e assegura garantias de privacidade robustas e formalmente comprovadas, distinguindo-se das soluções que se baseiam em testes práticos. Além disso, a Privacidade Diferencial também fornece controlos de privacidade personalizáveis e ajustáveis, permitindo que os utilizadores equilibrem o compromisso entre privacidade e utilidade de acordo com as suas necessidades.

Palavras-chave: Redes de Anonimato, Privacidade Diferencial, Comunicação Preservadora de Privacidade, Comunicação Encoberta, Ataques de Análise de Tráfego

Contents

List of Figures viii

1 Introduction 1
- 1.1 Motivation . . . 2
- 1.2 Problem . . . 2
- 1.3 Goals . . . 3
  - 1.3.1 Contributions . . . 3
- 1.4 Report Organization . . . 4

2 Related Work 5
- 2.1 Tor Network . . . 5
- 2.2 Anonymity Networks & Mixnets . . . 6
- 2.3 K-anonymity . . . 7
- 2.4 TorKameleon & Stream Covered Channels . . . 7
- 2.5 Differential Privacy . . . 7
- 2.6 Summary . . . 7

3 Elaboration 8
- 3.1 System Model . . . 8
- 3.2 Prototype Guidelines . . . 8
- 3.3 Validation . . . 8
- 3.4 Summary . . . 8

4 Workplan 9

Bibliography 10

List of Figures

1 Introduction

The internet has become an integral part of the daily life for most of the world's population. However, in some regions, individuals are denied the ability to freely share their stories or access vital information due to oppressive conditions and restrictions on freedom of speech. This oppression and restrictions are often enforced by totalitarian regimes that have vast resources and power to control the flow of information. [2, 46, 1]

As the demand for anonymity networks continues to expand, so does the necessity for robust privacy guarantees to safeguard secure and confidential communications. At the same time, authoritarian regimes intensify their efforts to restrict access exclusively to state-approved content, employing advanced measures to enforce their censorship policies and maintain control over information flow.

In order to enforce and control the network traffic, these regimes resort to the use of traffic analysis techniques to passively monitor the traffic and infer information about the users. Traffic analysis attacks are a significant threat to anonymity networks, as they can be used to infer users' activities and deny their privacy, as well as deanonymize them. [5, 42, 19] Some have demonstrated that advances in machine learning lead to more sophisticated and effective traffic correlation attacks, such as DeepCorr [26] and DeepCoFFEA [30], which leverage deep learning techniques to correlate traffic patterns and infer users' activities with high accuracy, in anonymity networks like Tor. [21]. Statistical Disclosure attacks also pose a threat to anonymity networks, as they identify sender-receiver pairs in anonymity networks. [10, 24] Website Fingerprinting attacks represent a major threat to anonymity networks, as they allow attackers to learn information about user's online activities, even on encrypted networks like Tor, by eavesdropping and collecting side-channel information from the network traffic, between the user and the entry node. With this information, the attacker can use machine learning techniques to infer the user's online activities, such as the websites visited, by analyzing the traffic patterns, with high accuracy. [36, 33, 7]

As these attacks become more sophisticated and effective, journalists, whistleblowers and any other person trying to express their freedom of speech and access information in these regions, are at risk of being identified and persecuted, leading to the decrease of this networks utility, mistrust and fear. Therefore, it is critical that anonymity networks

become more resilient to these attacks and provide stronger privacy guarantees to their users.

1.1 Motivation

Tor, one of the most popular anonymity networks, is an anonymous communication service based on the Onion Routing Protocol. [11] Tor is designed to protect users' privacy and security by routing their traffic through a network of volunteer-operated servers, encrypting it at each step, and ensuring that no single server knows both the source and destination of the traffic.

Anonymity networks remain vulnerable to attacks such as traffic analysis, which threatens the privacy and security of the users these networks are designed to protect. [5, 42, 19, 10, 24, 36, 33, 7] To address this challenge of circumventing censorship, researchers have proposed a range of solutions. Among these solutions, we discuss traffic encapsulation, fully encrypted protocols, and traffic shaping techniques.

Traffic encapsulation is a method that conceals data by embedding packets within other data formats or protocols, thereby obfuscating their true content. This technique leverages protocols, such as those for web browsing and video streaming, to disguise network traffic, making it indistinguishable. [38, 41, 4, 12, 16]

Traffic Shaping is another technique that aims to regulate data transmission to optimize network performance and to prevent timing and correlation attacks, contributing for privacy-prevention. Similarly, traffic shuffling is a technique that focus on reordering packets, with statistical or cryptographic techniques, to make it difficult for an attacker to correlate the traffic. [34, 22, 8, 32, 43]

Researchers have proposed various solutions. Hopper and Vasserman demonstrated that K-Anonymity is effective against traffic analysis attacks, such as Statistical Disclosure and long-term intersection attacks. [15] Nunes et al. proposed BriK, a Tor pluggable transport, that ensures uniform traffic patterns to make users indistinguishable. [29] Some work as Walkie-Talkie [39], Supersequence [40] and Glove [28] also use traffic shaping for defending against website fingerprinting attacks. Wright, Coull, and Monrose proposed Traffic Morphing [44], a traffic shaping technique that uses a Markov Chain to generate traffic patterns that are indistinguishable from the original traffic.

1.2 Problem

Although anonymity networks are designed to provide robust privacy and security, advancements in traffic analysis and correlation techniques have increasingly exposed vulnerabilities, reducing their reliability and trustworthiness for users—particularly in regions where pervasive surveillance by authoritarian regimes is a critical concern. These regimes actively invest in sophisticated surveillance methods and traffic manipulation strategies to maintain control over digital communication.

Tor, despite being one of the most widely used anonymity networks, is not immune to these challenges. It remains vulnerable to several attack vectors, including traffic correlation, congestion-based attacks, timing-based correlations, and fingerprinting techniques. [5, 21, 42, 19] Such vulnerabilities highlight the ongoing arms race between those striving to safeguard online anonymity and those seeking to undermine it through technical and infrastructural advancements.

The absence of standardized privacy metrics for evaluating privacy guarantees in a formal and rigorous manner raises concerns about the effectiveness of anonymizing networks like Tor. Research has demonstrated vulnerabilities in the Tor network to various attacks, including traffic analysis. For instance, Chakravarty et al. showed that statistical correlation techniques can be used to perform successful traffic analysis attacks on the Tor network. Additionally, Shi and Matsuura demonstrated that user privacy can be compromised through fingerprinting attacks.

1.3 Goals

The goal is to add differential privacy (DP) mechanisms into the Tor open-source project to preserve its existing security guarantees while enhancing its privacy protections. By incorporating DP, the aim is to provide quantifiable and adjustable privacy guarantees to users, addressing vulnerabilities to correlation attacks and improving resistance to traffic observability. Furthermore, the project seeks to analyze the trade-offs between privacy and utility within the Tor context and to evaluate the performance of the proposed solution compared to the original Tor network.

As stated in Section, it is important to ensure that anonymity networks are secure against threats like artificial intelligence and machine learning, and to address the lack of standardized privacy metrics to evaluate the performance of anonymity networks. We aim to create and implement a new algorithm that can be used in anonymity networks that leverages differential privacy to ensure privacy and security with higher performance than existing solutions. We also aim to evaluate the performance of the algorithm and compare it with other anonymity networks to prove that this solution is a viable alternative to usual anonymity networks solutions. To test the algorithm and its performance, we intend to realize correlation attacks on the existing networks and on our prototype and compare the results. To test performance, we will measure the latency and throughput of the algorithm and compare it with other anonymity networks.

1.3.1 Contributions

This thesis tries to address the problem of ensuring privacy and security in anonymity networks while maintaining high performance. In particular, the contributions of this thesis are:

1. Design a new Differential Privacy Based Algorithm for Privacy Preserved Communication in Anonymity Networks.
2. Implement the algorithm on top of the Tor network and provide a proof of concept.
3. Evaluate the performance of the algorithm and compare it with other anonymity networks; This contribution aims to prove if this solution is a viable alternative to usual anonymity networks solutions and if, as mention before, can allow users to switch from traditional to anonymity networks.

1.4 Report Organization

The remainder of this report is organized as follows: Chapter introduces anonymity networks and differential privacy, as well as some other related topics. The proposal of work is presented on Chapter, which provides some system model and guidelines for the development and discusses the proof method to verify the work. Finally, workplan is explained and addressed in Chapter

2

Related Work

2.1 Tor Network

Tor is a circuit-based low-latency anonymous communication service based on *onion routing* for asynchronous, loosely federated onion routers [Tor: The second-generation onion router]. This protocol is a low-latency anonymity network design that tries to protect users from correlation attacks. The overlay network relies on *Onion Routers* (ORs) that maintain TLS connection to every other onion router [11].

Onion Proxy are local software, run on the client and handle connections with the users' applications, fetch the current network information and the list of *Onion Routers* (ORs) and build circuits through the network. Each router maintains TLS connections to nodes they have been in contact recently and a pair of keys: long-term identity key and short-term onion key. Long-term keys are used to sign TLS certificates and its router descriptor. Short-term keys are used to decrypt requests, negotiate ephemeral keys and set up circuits.

Traffic travels the network through 512 bytes fixed-size cells with a header and a payload. The header contains the circuit identifier, the command to describe what to do with the payload. Based on the header's command, cells are either control cells or relay cells. Control cells are used to manage circuits and connections, while relay cells are used to carry data.

The sequences of ORs which traffic is routed through are called *circuits*. The user's Onion Proxy construct circuits, negotiating a symmetric key with each OR on the circuit, one hop at a time. Circuits incrementally extends, as well as torn down. Relay cells can only be sent after a circuit is established. Once established, relay cells are encapsulated in several layers of encryption, correspondent to each node in the circuit, with the pre-negotiated keys with each router. As it passes through each node, a layer of encryption is peeled off and forward to the next node. This ensures that only the last node knows the final destination of the data, and no single node knows both the source and destination of the data.

2.2 Anonymity Networks & Mixnets

Mix networks (Mixnets) were introduced by Chaum [6] as a public key cryptography based protocol designed to obscure the relationship between senders and receivers in a communication network, as well as its content, without the need for a universal trusted authority. Mixnets consist of a series of intermediary servers, called 'mixes', through which messages are decrypted, encrypted, randomly permuted and sent forward. To transmit messages, participants encapsulate their data within successive layers of encryption, with each layer corresponding to a specific mix in the network — a method conceptually akin to onion routing [13]. The length of the encrypted messages is proportional to the number of mixes. Upon receiving a message, each mix removes the outermost layer of encryption, processes the decrypted instructions, and forwards the partially decrypted message to the subsequent mix. This iterative procedure continues until the message arrives at the final mix, which removes the last layer of encryption and transmits the plaintext message to the intended recipient. By design, this system ensures that no single mix possesses complete knowledge of both the sender's identity and the recipient's address, thereby achieving a robust level of anonymity.

Various designs of mixnets have been proposed, based on Chaum work, addressing multiple security and performance issues. Flash Mixing, introduced by Jakobsson [17], focuses on achieving strong anonymity guarantees with reduced latency and computational overhead by shuffling messages, broadcasting the encrypted list to all mixes, who then together compute the output. Hybrid Mixnets [18, 31] efficiently combine public-key and symmetric-key cryptography. Real-time Mixnets [20] aim voice and data communication where continuous data streams have to be transmitted and provide low-latency communication, as long as a certain delay at the start of a connection is tolerable. Work on this field has focused both on security and privacy guarantees, as well as on performance and scalability, with the goal of providing a practical and efficient solution for anonymous communication. Mixmaster [25] allows the sender of a message to remain anonymous to the recipient. Mixminion [9] uses fixed sized messages and supports anonymous replies and ensures forward anonymity using link encryption between nodes. Onion Routing [13] followed a similar approach, but focused on low latency communication, where messages are encrypted in layers and decrypted by a chain of authorized nodes.

Tor [11] is one of the most popular low latency anonymity system and protects against sender-receiver message linking against a partially global adversary and ensures perfect forward secrecy, messages' integrity, and congestion control. However, Tor is vulnerable to traffic analysis attacks, in case the adversary is able to observe the entry and exit points of the network.

Mix-In-Place (MIP) [27] is a mixnet that uses a cascade of functions in a single proxy, instead of multiple intermediary nodes, to provide anonymity, and proving more resistant to traffic analysis attacks.

Vuvuzela [14] operates in rounds, leading to offline users' inability to receive messages

and all messages must transverse a single chain of relay servers. This design protects against both active and passive adversaries unless there is no honest mix node in the network.

Loopix [32]

Groove [3]

Stadium [37] is a horizontally scalable metadata-private messaging system design to provide strong privacy guarantees against adversaries capable of monitoring and manipulating traffic, by using techniques like differential privacy and parallel mixing. Unlike Vuvuzela, this mixnet system distributes the workload among hundreds of services.

Mixnets have been analyzed in terms of security and privacy guarantees, as well as performance and scalability. Zhu et al. revealed that flow-correlation attacks can be used to de-anonymize users in mixnets, in relation to the system parameterization, such as sample size, noise level, payload flow rate, and detection rate. [45]. Shmatikov and Wang demonstrated that mix networks are vulnerable to timing analysis, even with defenses in place, by the use of advanced statistical techniques, such as cross-correlation and machine learning, successfully de-anonymizing users. [0].

2.3 K-anonymity

2.4 TorKameleon & Stream Covered Channels

2.5 Differential Privacy

2.6 Summary

3

Elaboration

3.1 System Model

3.2 Prototype Guidelines

3.3 Validation

3.4 Summary

4

Workplan

Bibliography

[1] M. Alimardani and S. Milan. "The Internet as a Global/Local Site of Contestation: The Case of Iran". In: *Global Cultures of Contestion*. Ed. by D. T. Della and G. K. Thomas. Springer, 2018, pp. 171–192. URL: https://link.springer.com/chapter/10.1007/978-3-319-78889-7_9 (cit. on p. 1).

[2] S. Aryan, H. Aryan, and J. A. Halderman. "Internet Censorship in Iran: A First Look". In: *Proceedings of the 3rd USENIX Workshop on Free and Open Communications on the Internet (FOCI 13)*. USENIX Association, 2013. URL: https://www.usenix.org/conference/foci13/workshop-program/presentation/aryan (cit. on p. 1).

[3] L. Barman et al. "Groove: Flexible Metadata-Private Messaging". In: *16th USENIX Symposium on Operating Systems Design and Implementation (OSDI 22)*. Carlsbad, CA: USENIX Association, 2022-07, pp. 735–750. ISBN: 978-1-939133-28-1. URL: https://www.usenix.org/conference/osdi22/presentation/barman (cit. on p. 7).

[4] D. Barradas et al. "Poking a Hole in the Wall: Efficient Censorship-Resistant Internet Communications by Parasitizing on WebRTC". In: *Proceedings of the 2020 ACM SIGSAC Conference on Computer and Communications Security (CCS)*. ACM, 2020, pp. 35–48. DOI: 10.1145/3372297.3423345. URL: https://doi.org/10.1145/3372297.3423345 (cit. on p. 2).

[5] S. Chakravarty et al. "On the Effectiveness of Traffic Analysis against Anonymity Networks Using Flow Records". In: *Passive and Active Measurement*. Ed. by M. Faloutsos and A. Kuzmanovic. Cham: Springer International Publishing, 2014, pp. 247–257 (cit. on pp. 1–3).

[6] D. Chaum. "Untraceable Electronic Mail, Return Addresses and Digital Pseudonyms". In: *Secure Electronic Voting*. Ed. by D. A. Gritzalis. Boston, MA: Springer US, 2003, pp. 211–219. ISBN: 978-1-4615-0239-5. DOI: 10.1007/978-1-4615-0239-5_14. URL: https://doi.org/10.1007/978-1-4615-0239-5_14 (cit. on p. 6).

[7] G. Cherubin, R. Jansen, and C. Troncoso. "Online Website Fingerprinting: Evaluating Website Fingerprinting Attacks on Tor in the Real World". In: *31st USENIX Security Symposium (USENIX Security 22)*. Boston, MA: USENIX Association, 2022-08, pp. 753–770. ISBN: 978-1-939133-31-1. URL: https://www.usenix.org/conference/usenixsecurity22/presentation/cherubin (cit. on pp. 1, 2).

[8] H. Corrigan-Gibbs, D. Boneh, and D. Mazieres. "Riposte: An anonymous messaging system handling millions of users". In: *Proceedings of the IEEE Symposium on Security and Privacy (S&P)*. 2015, pp. 321–338 (cit. on p. 2).

[9] G. Danezis, R. Dingledine, and N. Mathewson. "Mixminion: design of a type III anonymous remailer protocol". In: *2003 Symposium on Security and Privacy, 2003*. 2003, pp. 2–15. DOI: 10.1109/SECPRI.2003.1199323 (cit. on p. 6).

[10] G. Danezis. "Statistical Disclosure Attacks". In: *Security and Privacy in the Age of Uncertainty*. Ed. by D. Gritzalis et al. Boston, MA: Springer US, 2003, pp. 421–426. ISBN: 978-0-387-35691-4 (cit. on pp. 1, 2).

[11] R. Dingledine, N. Mathewson, and P. Syverson. "Tor: The Second-Generation Onion Router". In: *Proceedings of the 13th USENIX Security Symposium*. USENIX Association, 2004, pp. 303–320. URL: https://www.usenix.org/conference/usenixsecurity04/tor-second-generation-onion-router (cit. on pp. 2, 5, 6).

[12] G. Figueira, D. Barradas, and N. Santos. "Stegozoa: Enhancing WebRTC Covert Channels with Video Steganography for Internet Censorship Circumvention". In: *Proceedings of the 2022 ACM on Asia Conference on Computer and Communications Security*. ASIA CCS '22. Nagasaki, Japan: Association for Computing Machinery, 2022, 1154–1167. ISBN: 9781450391405. DOI: 10.1145/3488932.3517419. URL: https://doi.org/10.1145/3488932.3517419 (cit. on p. 2).

[13] D. Goldschlag, M. Reed, and P. Syverson. "Hiding Routing Information". In: 1996-08. ISBN: 978-3-540-61996-3. DOI: 10.1007/3-540-61996-8_37 (cit. on p. 6).

[14] J. van den Hooff et al. "Vuvuzela: scalable private messaging resistant to traffic analysis". In: *Proceedings of the 25th Symposium on Operating Systems Principles*. SOSP '15. Monterey, California: Association for Computing Machinery, 2015, 137–152. ISBN: 9781450338349. DOI: 10.1145/2815400.2815417. URL: https://doi.org/10.1145/2815400.2815417 (cit. on p. 6).

[15] N. Hopper and E. Y. Vasserman. "On the effectiveness of k;-anonymity against traffic analysis and surveillance". In: *Proceedings of the 5th ACM Workshop on Privacy in Electronic Society*. WPES '06. Alexandria, Virginia, USA: Association for Computing Machinery, 2006, 9–18. ISBN: 1595935568. DOI: 10.1145/1179601.1179604. URL: https://doi.org/10.1145/1179601.1179604 (cit. on p. 2).

[16] A. Houmansadr et al. "IP over Voice-over-IP for censorship circumvention". In: *CoRR* abs/1207.2683 (2012). arXiv: 1207.2683. URL: http://arxiv.org/abs/1207.2683 (cit. on p. 2).

[17] M. Jakobsson. "Flash mixing". In: *Proceedings of the Eighteenth Annual ACM Symposium on Principles of Distributed Computing*. PODC '99. Atlanta, Georgia, USA: Association for Computing Machinery, 1999, 83–89. ISBN: 1581130996. DOI: 10.1145/301308.301333. URL: https://doi.org/10.1145/301308.301333 (cit. on p. 6).

[18] M. Jakobsson and A. Juels. "An optimally robust hybrid mix network". In: *Proceedings of the Twentieth Annual ACM Symposium on Principles of Distributed Computing*. PODC '01. Newport, Rhode Island, USA: Association for Computing Machinery, 2001, 284–292. ISBN: 1581133839. DOI: 10.1145/383962.384046. URL: https://doi.org/10.1145/383962.384046 (cit. on p. 6).

[19] R. Jansen, T. Vaidya, and M. Sherr. "Point Break: A Study of Bandwidth Denial-of-Service Attacks against Tor". In: *28th USENIX Security Symposium (USENIX Security 19)*. Santa Clara, CA: USENIX Association, 2019-08, pp. 1823–1840. ISBN: 978-1-939133-06-9. URL: https://www.usenix.org/conference/usenixsecurity19/presentation/jansen (cit. on pp. 1–3).

[20] A. Jerichow et al. "Real-time mixes: a bandwidth-efficient anonymity protocol". In: *IEEE J.Sel. A. Commun.* 16.4 (2006-09), 495–509. ISSN: 0733-8716. DOI: 10.1109/49.668973. URL: https://doi.org/10.1109/49.668973 (cit. on p. 6).

[21] A. Johnson et al. "Users Get Routed: Traffic Correlation on Tor by Realistic Adversaries". In: *Proceedings of the ACM SIGSAC Conference on Computer and Communications Security (CCS)*. ACM, 2013. DOI: 10.1145/2508859.2516651. URL: https://doi.org/10.1145/2508859.2516651 (cit. on pp. 1, 3).

[22] D. Lazar, Y. Gilad, and N. Zeldovich. "Karaoke: Distributed private messaging immune to passive traffic analysis". In: *Proceedings of the 13th USENIX Symposium on Operating Systems Design and Implementation (OSDI)*. 2018, pp. 711–725 (cit. on p. 2).

[23] J. M. Lourenço. *The NOVAthesis LaTeX Template User's Manual*. NOVA University Lisbon. 2021. URL: https://github.com/joaomlourenco/novathesis/raw/main/template.pdf (cit. on p. i).

[24] N. Mathewson and R. Dingledine. "Practical Traffic Analysis: Extending and Resisting Statistical Disclosure". In: *Privacy Enhancing Technologies*. Ed. by D. Martin and A. Serjantov. Berlin, Heidelberg: Springer Berlin Heidelberg, 2005, pp. 17–34. ISBN: 978-3-540-31960-3 (cit. on pp. 1, 2).

[25] U. Moeller. "Mixmaster Protocol Version 2". In: 2004. URL: https://api.semanticscholar.org/CorpusID:215824742 (cit. on p. 6).

[26] M. Nasr, A. Bahramali, and A. Houmansadr. "DeepCorr: Strong Flow Correlation Attacks on Tor Using Deep Learning". In: *Proceedings of the ACM SIGSAC Conference on Computer and Communications Security*. Toronto, ON, Canada: ACM, 2018, pp. 1962–1976. DOI: 10.1145/3243734.3243858 (cit. on p. 1).

[27] N. Nipane, I. Dacosta, and P. Traynor. “"Mix-in-Place" anonymous networking using secure function evaluation”. In: *Proceedings of the 27th Annual Computer Security Applications Conference*. ACSAC ’11. Orlando, Florida, USA: Association for Computing Machinery, 2011, 63–72. ISBN: 9781450306720. DOI: 10.1145/2076732.2076742. URL: https://doi.org/10.1145/2076732.2076742 (cit. on p. 6).

[28] R. Nithyanand, X. Cai, and R. Johnson. “Glove: A Bespoke Website Fingerprinting Defense”. In: *Proceedings of the 13th Workshop on Privacy in the Electronic Society*. WPES ’14. Scottsdale, Arizona, USA: Association for Computing Machinery, 2014, 131–134. ISBN: 9781450331487. DOI: 10.1145/2665943.2665950. URL: https://doi.org/10.1145/2665943.2665950 (cit. on p. 2).

[29] V. Nunes et al. “Enhancing the Unlinkability of Circuit-Based Anonymous Communications with k-Funnels”. In: *Proceedings of the ACM on Networking* 1.CoNEXT3 (2023), Article 18. DOI: 10.1145/3629140. URL: https://doi.org/10.1145/3629140 (cit. on p. 2).

[30] S. E. Oh et al. “DeepCoFFEA: Improved Flow Correlation Attacks on Tor via Metric Learning and Amplification”. In: *Proceedings of the IEEE Symposium on Security and Privacy*. San Francisco, CA, USA: IEEE, 2022, pp. 1915–1932. DOI: 10.1109/SP46214.2022.9833742 (cit. on p. 1).

[31] M. Ohkubo and M. Abe. “A Length-Invariant Hybrid Mix”. In: *Advances in Cryptology - ASIACRYPT 2000, 6th International Conference on the Theory and Application of Cryptology and Information Security, Kyoto, Japan, December 3-7, 2000, Proceedings*. Vol. 1976. Lecture Notes in Computer Science. Springer, 2000, pp. 178–191. DOI: 10.1007/3-540-44448-3_14 (cit. on p. 6).

[32] A. M. Piotrowska et al. “The Loopix Anonymity System”. In: *26th USENIX Security Symposium (USENIX Security 17)*. Vancouver, BC: USENIX Association, 2017-08, pp. 1199–1216. ISBN: 978-1-931971-40-9. URL: https://www.usenix.org/conference/usenixsecurity17/technical-sessions/presentation/piotrowska (cit. on pp. 2, 7).

[33] M. S. Rahman et al. “Tik-Tok: The Utility of Packet Timing in Website Fingerprinting Attacks”. In: *Proceedings on Privacy Enhancing Technologies* 2020.3 (2020), pp. 5–24. DOI: 10.2478/popets-2020-0032 (cit. on pp. 1, 2).

[34] T. Shen et al. “ DAENet: Making Strong Anonymity Scale in a Fully Decentralized Network ”. In: *IEEE Transactions on Dependable and Secure Computing* 19.04 (2022-07), pp. 2286–2303. ISSN: 1941-0018. DOI: 10.1109/TDSC.2021.3052831. URL: https://doi.ieeecomputersociety.org/10.1109/TDSC.2021.3052831 (cit. on p. 2).

[35] Y. Shi and K. Matsuura. “Fingerprinting Attack on the Tor Anonymity System”. In: *Information and Communications Security*. Ed. by S. Qing, C. J. Mitchell, and G. Wang. Berlin, Heidelberg: Springer Berlin Heidelberg, 2009, pp. 425–438 (cit. on p. 3).

[0] V. Shmatikov and M.-H. Wang. "Timing Analysis in Low-Latency Mix Networks: Attacks and Defenses". In: vol. 4189. 2006-01, pp. 18–33. ISBN: 978-3-540-44601-9. DOI: 10.1007/11863908_2 (cit. on p. 7).

[36] P. Sirinam et al. "Deep Fingerprinting: Undermining Website Fingerprinting Defenses with Deep Learning". In: *Proceedings of the 2018 ACM SIGSAC Conference on Computer and Communications Security*. CCS '18. Toronto, Canada: Association for Computing Machinery, 2018, 1928–1943. ISBN: 9781450356930. DOI: 10.1145/3243734.3243768. URL: https://doi.org/10.1145/3243734.3243768 (cit. on pp. 1, 2).

[37] N. Tyagi et al. "Stadium: A Distributed Metadata-Private Messaging System". In: *Proceedings of the 26th Symposium on Operating Systems Principles*. SOSP '17. Shanghai, China: Association for Computing Machinery, 2017, 423–440. ISBN: 9781450350853. DOI: 10.1145/3132747.3132783. URL: https://doi.org/10.1145/3132747.3132783 (cit. on p. 7).

[38] A. Vilalonga, J. S. Resende, and H. Domingos. *TorKameleon: Improving Tor's Censorship Resistance with K-anonymization and Media-based Covert Channels*. 2023. arXiv: 2303.17544 [cs.CR]. URL: https://arxiv.org/abs/2303.17544 (cit. on p. 2).

[39] T. Wang and I. Goldberg. "Walkie-Talkie: An Efficient Defense Against Passive Website Fingerprinting Attacks". In: *26th USENIX Security Symposium (USENIX Security 17)*. Vancouver, BC: USENIX Association, 2017-08, pp. 1375–1390. ISBN: 978-1-931971-40-9. URL: https://www.usenix.org/conference/usenixsecurity17/technical-sessions/presentation/wang-tao (cit. on p. 2).

[40] T. Wang et al. "Effective Attacks and Provable Defenses for Website Fingerprinting". In: *Proceedings of the 23rd USENIX Security Symposium*. San Diego, CA, USA: USENIX Association, 2014, pp. 143–156. URL: https://www.usenix.org/conference/usenixsecurity14/technical-sessions/presentation/wang_tao (cit. on p. 2).

[41] Z. Weinberg et al. "StegoTorus: A camouflage proxy for the Tor anonymity system". In: 2012-10, pp. 109–120. DOI: 10.1145/2382196.2382211 (cit. on p. 2).

[42] P. Winter and S. Lindskog. "How the Great Firewall of China is Blocking Tor". In: *Proceedings of the 2nd USENIX Workshop on Free and Open Communications on the Internet (FOCI)*. USENIX Association, 2012. URL: https://www.usenix.org/conference/foci12/workshop-program/presentation/winter (cit. on pp. 1–3).

[43] D. I. Wolinsky et al. "Dissent in Numbers: Making Strong Anonymity Scale". In: *10th USENIX Symposium on Operating Systems Design and Implementation (OSDI 12)*. Hollywood, CA: USENIX Association, 2012-10, pp. 179–182. ISBN: 978-1-931971-96-6. URL: https://www.usenix.org/conference/osdi12/technical-sessions/presentation/wolinsky (cit. on p. 2).

[44] C. V. Wright, S. E. Coull, and F. Monrose. "Traffic Morphing: An Efficient Defense Against Statistical Traffic Analysis". In: *Network and Distributed System Security Symposium*. 2009. URL: https://api.semanticscholar.org/CorpusID:2562331 (cit. on p. 2).

[45] Y. Zhu et al. "Anonymity analysis of mix networks against flow-correlation attacks". In: *GLOBECOM '05. IEEE Global Telecommunications Conference, 2005.* Vol. 3. 2005, 5 pp.–. DOI: 10.1109/GLOCOM.2005.1577959 (cit. on p. 7).

[46] J. L. Zittrain et al. "The Shifting Landscape of Global Internet Censorship". In: Accessed: 2024-11-17. Berkman Klein Center for Internet & Society, 2017. URL: https://www.ssrn.com/abstract=2993485 (cit. on p. 1).